



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 226532
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
4/8/2025

Sold To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140	Ship To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140
Customer ID	34217		
TFA Sales Order	326358\2-1		
Purchase Order	662682080		

TFA Part ID	55214C	Heat Code	A2794A
Description	10 X 4 XH TEE Y-52 w/CAP	Grade	Y52
Quantity	18	Raw Material Heat	045179
Customer Part ID	ZFIT	Process	Y52-3BX
		Mill	TPCO

Austenitized @:	1750°	Tempered @:	1100°	Stress Relieved @:		Medium:	WATER
Held For:	60 mins/min	Held For:	60 mins/min	Held For:	mins/min		

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.15	1.13	0.012	0.003	0.26	0.02	0.01	0.02	0.02	<0.01	<0.01	0.0005	0.01	0.02				0.34

Product Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.14	1.11	0.010	0.001	0.28	0.01	0.01	0.01	0.03	<0.01	<0.01	<0.0010	0.01	0.02				0.32

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1	Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	62100 (psi)	80400 (psi)	32.00	73.10	2"	201			TRANSVERSE	FLAT (.501" Width)

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)				% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg			

Notes/Comments

1. THIS MATERIAL WAS AUSTENITIZED AT 1750F, WATER QUENCHED, AND TEMPERED AT 1100F. 2. THIS MATERIAL IS OF SEAMLESS MANUFACTURE.

1) MATERIAL MEETS MSS-SP-75 - 2019.

2) MATERIAL MEETS NACE MR-01-75/ISO 15156, REGIONS 1, 2 AND 3 - 2015 EDITION, ALSO MEETS NACE MR-0103 2015 EDITION.

3) MANUFACTURED IN AN ISO 9001 CERTIFIED FACILITY-CERTIFICATE #30248.

4) MATERIAL UNDER 16 NPS IS NOT IMPACT TESTED UNLESS ORDERED AS Y65 OR HIGHER YIELDS OR SUPPLEMENTARY REQUIREMENT (SR24) IS INVOKED PER MSS.

*UB = Unbroken Test

Heat: A2794A Grade: Y52

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Scot Marshall, Manager Quality Assurance

The recording of fictitious entries on this document and or its fraudulent misuse may be punishable as a felony under Federal Statute.

Version 2.13.32.14

Job Number

254636



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Quantity	18	Raw Material Heat	045179
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		Mill	TPCO

TEE RUN IS STRAIGHT BORED.

TEE RUN MAINTAINS AN INSIDE DIAMETER NO LESS THAN 98% OF THE SPECIFIED NOMINAL INSIDE DIAMETER.

TEE RUN IS CAPABLE OF RECEIVING A PIGGING DEVICE SUITED TO TRAVEL THROUGH A FITTING WITH AN INSIDE DIAMETER NOT LESS THAN 98% OF THE SPECIFIED NOMINAL INSIDE DIAMETER.

Material Formed/Forged At: Tube Forgings of
America, Inc, Portland OR

Material Heat Treated At: Met-Tek,
Clackamas OR

TFA FITTINGS ARE MANUFACTURED IN PORTLAND, OR, USA. FOR THIS HEAT CODE, RAW MATERIAL SOURCE IS CHINA

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